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Management of Pharyngeal Perforation with Cervical Spine Injury Following Blunt Trauma on Electric Vehicle Accident

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Data Collection B
Statistical Analysis C
Data Interpretation D
Manuscript Preparation E
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Patient: Female, 58-year-old
Final Diagnosis: Cervical ligamentous complex rupture • pharyngeal laceration • vertebral plate fracture
Symptoms: Dyspnea • neck pain • neck swelling • shock
Clinical Procedure: —
Specialty: Orthopedics and Traumatology • Otolaryngology

Objective: Rare disease
Background: Cases of pharyngeal perforation (PP) associated with cervical spine injury following blunt trauma are extremely rare.
Case Report: A 58-year-old woman was hit by a car while riding an electric bicycle. She presented with neck pain and dyspnea on admission to the hospital. The CT scan showed suspected fractures of the 5th and 6th cervical vertebrae without evidence of cervical emphysema. Serial blood tests revealed a rapidly decreasing hemoglobin level. An urgent surgical exploration of the neck was performed. Intraoperative findings included complete tears of the anterior and posterior longitudinal ligaments of the cervical spine, rupture of the 5th and 6th cervical intervertebral discs, and a longitudinal laceration of the posterior pharyngeal wall. The patient underwent surgical management, which included PP repair and cervical spine infusion via an anterior cervical approach. The buccopharyngeal fascia was sutured to the prevertebral fascia to seal the connection between the PP and the prevertebral space. The patient recovered well after surgery. The 3-month postoperative follow-up showed a stable cervical spine and good healing of the PP.
Conclusions: We present the first case of PP associated with severe cervical spine injury following blunt trauma. Direct laryngoscopy is essential to establish the presence of PP and to assess the size and location of the injury. In this rare case, suturing the buccopharyngeal fascia and prevertebral fascia to isolate the PP and spinal implant was effective in preventing implant infection and surgical failure potentially caused by salivary leakage.

Keywords: Neck Injuries • Pharynx • Cervical Vertebrae • Lacerations

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Introduction

The oropharynx and hypopharynx are particularly susceptible to traumatic injuries, which, if not promptly identified and treated, can lead to severe complications such as emphysema, posterior pharyngeal space abscesses, and life-threatening mediastinal infections due to the involvement of multiple neck spaces. Early detection and management of pharyngeal trauma are therefore critical to prevent these potentially devastating outcomes.

Iatrogenic causes are the most common cause of pharyngeal injuries, followed by trauma from endoscopy, intubation, or anterior cervical spinal surgery [1-3]. In a single-center retrospective study, iatrogenic factors accounted for 68% of pharyngeal and esophageal trauma, while 14% were due to foreign bodies, 11% occurred after discectomy and fusion, and 4% resulted from severe vomiting and penetrating injuries [4]. Rare causes of pharyngeal perforation include pneumatic pressure injuries and choking [5,6]. Berry et al estimated that the incidence of pharyngoesophageal perforation due to blunt trauma was less than 2% [7], a figure corroborated by Barkovich et al, who noted that despite the high volume of endoscopic and cervical spine procedures performed over the past 50 years, the incidence of blunt trauma-induced perforation remains below 2% [8].

Cervical spine disorders caused by motor vehicle accidents, commonly referred to as whiplash injuries, were first described by Crowe in 1928 [9]. These injuries typically involve hyperextension and hyperflexion of the neck. The Québec Task Force (QTF) on Whiplash-Associated Disorders developed a classification system with 5 grades [10], ranging from QTF 0 ("no neck complaints or physical signs") to QTF 4 ("neck complaints with fracture or dislocation"). Most cases fall into the QTF 0-3 categories; QTF 4 injuries are exceptionally rare.

The present article presents the first reported case of pharyngeal perforation associated with severe whiplash injury (QTF 4) due to blunt trauma. By reviewing the relevant literature and exploring potential mechanisms of injury, this case highlights the importance of considering pharyngeal perforation in the context of severe cervical spine trauma, even in the absence of penetrating injuries. The findings emphasize the need for increased clinical vigilance and a multidisciplinary approach to the management of such complex cases, which may inform future diagnostic and therapeutic strategies. This case also contributes to the limited knowledge of rare complications of blunt neck trauma, providing insights that may improve patient outcomes and guide further research in this area.

Case Report

Case Presentation

A 58-year-old female patient was admitted to the Emergency Department of the Third Division General Hospital in Xinjiang after being struck from behind by a car while riding an electric bicycle on May 17, 2024. Upon arrival, she presented with neck pain and dyspnea. Initial clinical examination revealed a heart rate of 105 beats per minute (bpm), blood pressure of 89/52 mmHg, dry and cold skin, pale lips and mouth, and fresh blood in the oral and nasal cavities. Notable neck swelling was observed (**Figure 1A**). Blood tests showed a hemoglobin (Hb) level of 10.0 g/dL.

Considering the possibility of a cervical spine injury, the patient was immediately fitted with a cervical collar to protect the cervical spine. Given the patient's signs of shock and severe neck swelling, which posed a risk of asphyxiation, the emergency team immediately performed endotracheal intubation.

A whole-body CT scan was performed, which revealed suspected fractures of the 5th cervical vertebral plate and the 6th cervical transverse process, as well as widening of the intervertebral space between C5 and C6 (**Figure 1B**). The scan also showed thickening of the cervical soft tissues and the upper mediastinal septum, but no evidence of cervical emphysema. Post-CT, the patient's Hb level dropped significantly to 6.3 g/dL, suggesting the presence of active bleeding.

Given the patient's deteriorating condition, a multidisciplinary team of specialists from the intensive care unit (ICU), emergency department, otolaryngology, and spine surgery departments convened to discuss the case. The team concluded that the patient's persistent and worsening neck swelling, combined with the presence of fresh blood in the mouth and nasal cavity and the CT findings, pointed to bleeding from a ruptured blood vessel in the neck as the primary cause of hemorrhagic shock. The possibility of concomitant pharyngeal perforation was also considered, but could not be confirmed without further investigation. Due to the critical nature of the patient's condition and the need for urgent intervention, the decision was made to proceed with immediate surgical exploration of the neck.

Surgical Intervention

Intraoperative findings included extensive contusion and bruising of the soft tissues of the neck, a significant amount of blood clots in the prevertebral space, and complete rupture of the bilateral cervical longissimus muscles, as well as the anterior and posterior longitudinal ligaments. Additionally, there was a rupture of the annulus fibrosus of the C5 and C6 intervertebral discs, detachment of the endplates (**Figure 1C**), and

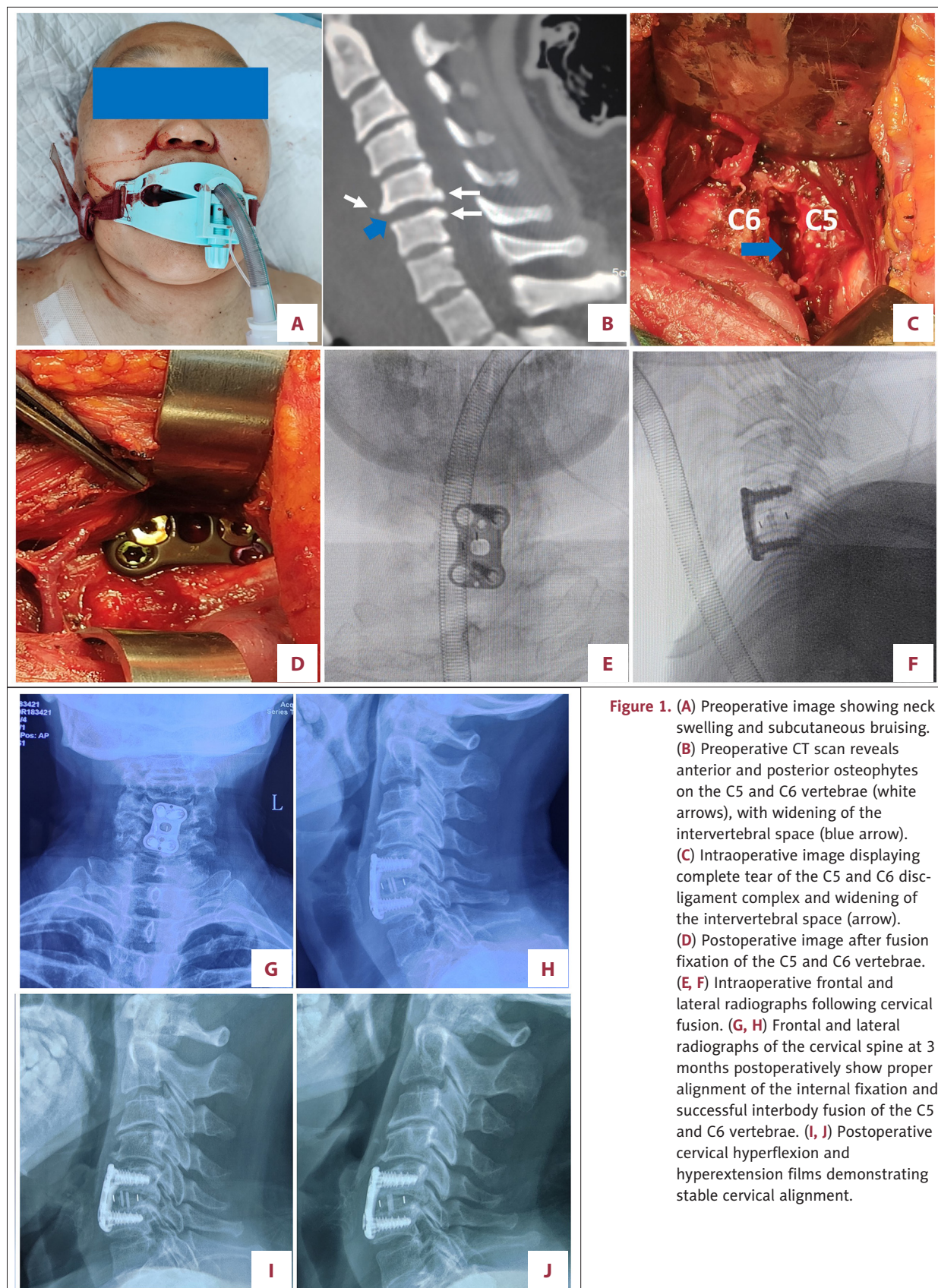


Figure 1. (A) Preoperative image showing neck swelling and subcutaneous bruising. (B) Preoperative CT scan reveals anterior and posterior osteophytes on the C5 and C6 vertebrae (white arrows), with widening of the intervertebral space (blue arrow). (C) Intraoperative image displaying complete tear of the C5 and C6 disc-ligament complex and widening of the intervertebral space (arrow). (D) Postoperative image after fusion fixation of the C5 and C6 vertebrae. (E, F) Intraoperative frontal and lateral radiographs following cervical fusion. (G, H) Frontal and lateral radiographs of the cervical spine at 3 months postoperatively show proper alignment of the internal fixation and successful interbody fusion of the C5 and C6 vertebrae. (I, J) Postoperative cervical hyperflexion and hyperextension films demonstrating stable cervical alignment.

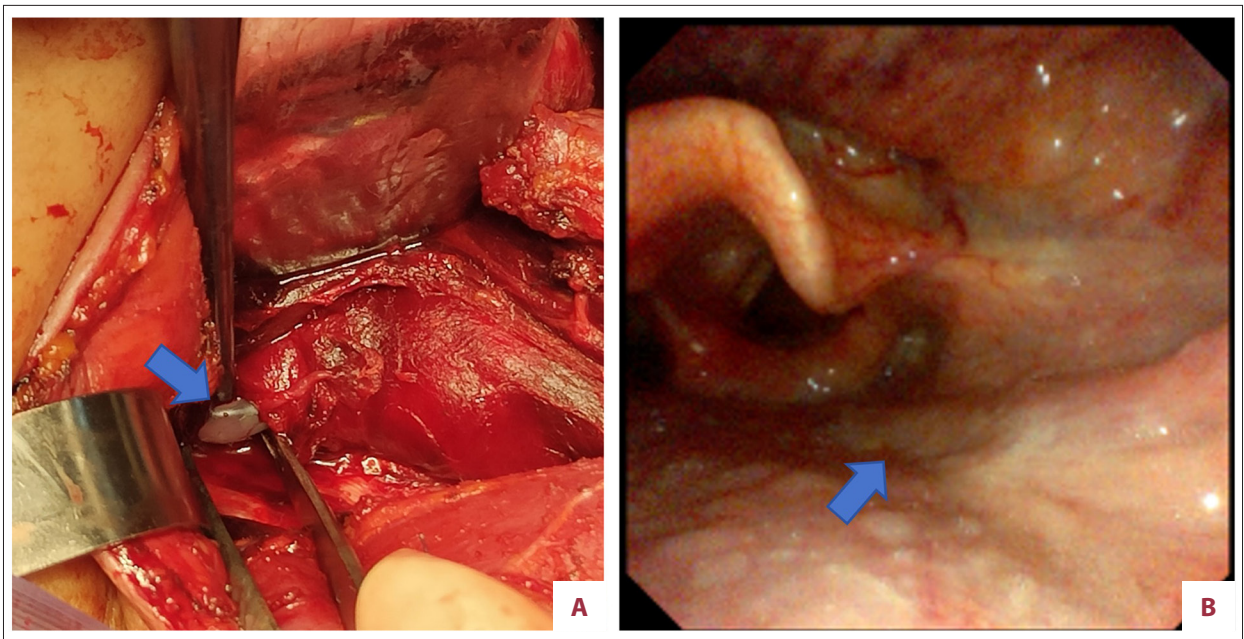


Figure 2. (A) View of the pharyngeal perforation through the neck via the gastric tube (arrow). (B) Fiberoptic laryngoscopy at 3 months postoperatively reveals a longitudinal scar (arrow) on the posterior pharyngeal wall.

significant hemorrhage in the anterior intervertebral space and the C5-6 intervertebral space.

To stabilize the cervical spine, a 14×6 mm intervertebral fusion device and artificial bone were implanted, and an external anterior cervical locking plate (Beijing Keyi Bone 24-mm plate with 45×16×3 mm and 40×16×1 mm screws) was applied (Figure 1D-1F). Suspended pharyngoscopy revealed a longitudinal laceration in the mucosa and constrictor muscle on the right side of the posterior pharyngeal wall, approximately 3 cm in length, just below the base of the tongue. This laceration communicated with the prevertebral space and was identified using a gastrostomy tube guide from the cervical approach (Figure 2A). The laceration was closed with 2-0 Vicryl sutures, and the buccopharyngeal fascia was sutured to the prevertebral fascia below the pharyngeal laceration to seal the connection between the perforation and the prevertebral space.

Postoperative Course

Postoperative drainage initially consisted of light red exudate, which changed to clear transparent fluid by the third postoperative day. Analysis of the fluid revealed a glucose level of 5.7 mmol/L, confirming cerebrospinal fluid (CSF) leakage. The vacuum drainage was replaced with a bag drainage, and the CSF leakage resolved after 3 days. The drain was removed 1 week postoperatively. The patient was extubated 2 days postoperatively, but developed respiratory instability 1 day after extubation. A CT scan revealed delayed bilateral cerebellar hemisphere and occipital injuries with minor hemorrhage.

The patient was therefore reintubated. She developed aspiration pneumonia during recovery but was successfully treated with a 2-week course of antibiotics and discharged thereafter.

Follow-Up

At the 3-month postoperative follow-up, pharyngoscopy showed longitudinal scarring at the junction of the right lateral and posterior pharyngeal walls (Figure 2B). Cervical radiographs demonstrated stable positioning of the internal fixation implant (Figure 1G, 1H), and flexion-extension radiographs revealed no instability at the injured segments (Figure 1I, 1J).

Discussion

Pharyngeal perforation resulting from blunt trauma is a rare but potentially life-threatening condition. The present case report highlights a unique instance of pharyngeal laceration associated with cervical ligamentous complex rupture and vertebral plate fracture, a combination not previously documented in the literature. By comparing this case to existing studies and exploring its peculiarities, we aim to contribute to the understanding of diagnostic and management strategies for such injuries.

Barkovich et al [8] conducted a retrospective review of 29 cases of pharyngeal perforation caused by blunt trauma, reported between 1964 and 2021. They found that most perforations occurred in the posterior pharyngeal wall, a region thought to

be particularly vulnerable due to its anatomic characteristics. Notably, 52% of these injuries resulted from motor vehicle accidents, underscoring the role of high-impact trauma in such cases [8]. Our literature review identified only 1 case of pharyngeal perforation due to blunt trauma in a female patient reported in the English literature since 2021 [11]. Wolf et al described a unique case of pharyngeal perforation associated with a cervical spine fracture from blunt trauma, which healed conservatively due to the small size of the perforation [12]. In contrast, our case involved a more severe injury pattern, including cervical ligamentous complex rupture and vertebral plate fracture, which required urgent surgical intervention.

Cervical emphysema is a common indicator of pharyngeal perforation and can be detected by plain neck films or CT scans [8]. However, in this case, the CT scan revealed only a diffuse hematoma with no evidence of emphysema, highlighting the limitations of imaging in certain scenarios. Fiberoptic pharyngoscopy is another diagnostic tool that can localize perforations, but mucosal folds may obscure the injury. In our patient, the severity and urgency of the prevertebral hemorrhage precluded preoperative fiberoptic laryngoscopy. Consequently, direct laryngoscopy was performed intraoperatively, which successfully identified a 3 cm longitudinal laceration in the posterior pharyngeal wall communicating with the prevertebral space. This decision highlights the importance of intraoperative diagnostic techniques when preoperative imaging is inconclusive.

The patient's mechanism of injury – being struck from behind while riding an electric bicycle – resulting in neck hyperextension and hyperflexion, occurred in the context of preexisting cervical spondylosis, which likely contributed to the severity of her injuries. Cervical spondylosis, characterized by osteophyte formation at the C5-C6 vertebrae (**Figure 1B**), may have predisposed her to ligamentous and vertebral damage under traumatic stress. The extreme forces exerted on the neck resulted in a complete tear of the cervical disc-ligamentous complex and a vertebral plate fracture, while the posterior pharyngeal wall, which is inherently fragile, sustained a laceration. This combination of injuries is extremely rare and underscores the importance of considering pre-existing spinal conditions in trauma management.

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Emergency surgery was imperative due to ongoing bleeding and hemorrhagic shock with the risk of secondary cervical cord injury. We tried to place a thin shoulder pad for better exposure prior to surgery. But when we tried, the bleeding became more severe. Therefore, we elevated the patient's head and kept the neck in a slight flexion to prevent further injury and bleeding. Anterior cervical immobilization was selected, and follow-up at 3 months confirmed that this approach provided adequate stabilization without the need for more invasive anterior-posterior fusion.

The length of the pharyngeal laceration and its communication with the prevertebral space made endoscopic repair impractical. Instead, a cervical approach with direct visualization was employed to suture the laceration and isolate the pharyngeal fistula from the cervical implant by approximating the buccopharyngeal and prevertebral fasciae. This technique not only facilitated healing but also minimized the risk of implant infection. The patient's uneventful recovery validated the appropriateness of these intraoperative decisions.

Conclusions

This case underscores the critical role of direct pharyngoscopy in the diagnosis of suspected pharyngeal perforations, especially when imaging findings are inconclusive. It also highlights the importance of protecting the cervical spine during management to prevent secondary spinal cord injury. Furthermore, in cases where a cervical implant communicates with a pharyngeal perforation, suturing the buccopharyngeal fascia to the prevertebral fascia is a valuable technique to isolate the fistula, prevent infection, and promote successful healing. These findings may guide the management of similar rare and complex injuries in the future.

Declaration of Figures' Authenticity

All figures submitted have been created by the authors who confirm that the images are original with no duplication and have not been previously published in whole or in part.

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